MILITARY SPECIFICATION SHEET

PLUG ASSEMBLIES, RIGHT ANGLE

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the plug assemblies described herein shall consist of this document and the latest issue of MIL-C-24217.

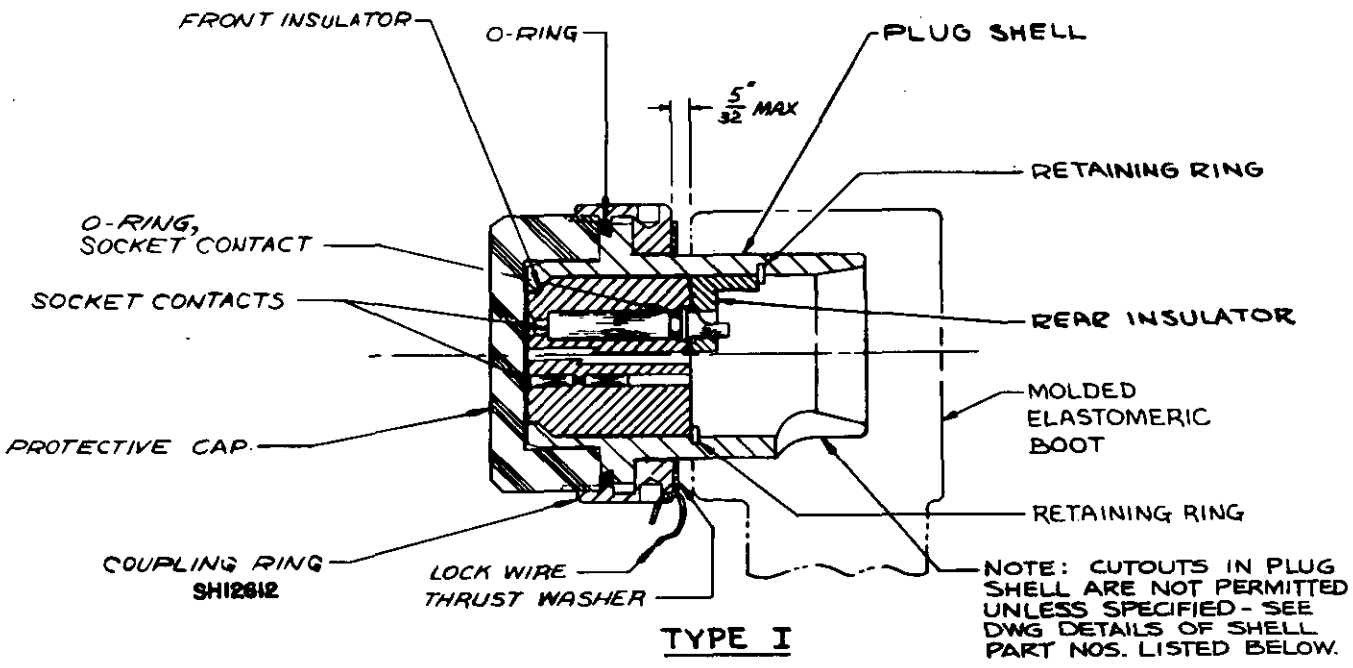


FIGURE 1. Configuration, type I.

FSC 5935

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TABLE I. Part numbers for right angle plug assemblies.

PLUG ASSEMBLY NUMBER	SOCKET CONTACT	O-RING, SOC. CONTACT AS-568A	NO. AND SIZE OF CONTACTS	PLUG SHELL NUMBER	COUPLING RING NUMBER	FRONT INSULATOR NUMBER	REAR INSULATOR NUMBER	PROT CAP NUMBER	THRUST WASHER NUMBER	RET. RING SPIROLOX NUMBER	SHELL O-RING AS-568A
M24217/8-001	MS17804-16-16	———	3 NO.16	M24217/7-024	M24217/7-039	M24217/7-050	———	M24217/7-078	M24217/7-093	UR-62S	-117
M24217/8-002	MS17804-16-20										
M24217/8-003	MS17804-16-16	———	5 NO.16	M24217/7-025	M24217/7-040	M24217/7-051	———	M24217/7-079	M24217/7-094	UR-75S	-122
M24217/8-004	MS17804-16-20										
M24217/8-005	MS17804-16-16	———	9 NO.16	M24217/7-027	M24217/7-041	M24217/7-052	———	M24217/7-080	M24217/7-095	UR-87S	-126
M24217/8-006	MS17804-16-20										
M24217/8-007	MS17804-16-16	———	14 NO.16	M24217/8-035	M24217/7-042	M24217/7-053	———	M24217/7-081	M24217/7-096	UR-112S	-132
M24217/8-008	MS17804-16-20										
M24217/8-009	MS17804-16-16	———	24NO16	M24217/8-036	M24217/7-043	M24217/7-054	———	M24217/7-082	M24217/7-097	UR-150S	-138
M24217/8-010	MS17804-16-20										
M24217/8-011	MS17804-16-16	———	37NO.16	M24217/8-037	M24217/7-044	M24217/7-055	———	M24217/7-083	M24217/7-098	UR-181S	-145
M24217/8-012	MS17804-16-20										
M24217/8-013	MS17804-16-16	———	48 NO.16	M24217/8-038	M24217/7-045	M24217/7-056	———	M24217/7-084	M24217/7-099	UR-225S	-151
M24217/8-014	MS17804-16-20										
M24217/8-015	M24217/7-089	-004	3 NO.12	M24217/7-032	M24217/7-046	M24217/7-057	M24217/7-071	M24217/7-085	M24217/7-095	UR-87S	-126
M24217/8-016	M24217/7-089	-004	5 NO.12	M24217/8-039	M24217/7-047	M24217/7-058	M24217/8-058	M24217/7-086	M24217/7-09	UR-112S	-132
M24217/8-017	M24217/7-089	-004	9 NO.12	M24217/8-040	M24217/7-048	M24217/7-059	M24217/8-059	M24217/7-087	M24217/7-097	UR-150S	-138
M24217/8-018	M24217/7-090	-007	3 NO.8	M24217/8-041	M24217/7-047	M24217/7-060	M24217/8-060	M24217/7-086	M24217/7-096	UR-112S	-132
M24217/8-019	M24217/7-091	-010	5 NO.4	M24217/8-040	M24217/7-048	M24217/7-061	M24217/8-061	M24217/7-087	M24217/7-097	UR-150S	-138
M24217/8-020	M24217/7-089	-004	3 NO.12 WITH 3 NO.8	M24217/8-040	M24217/7-048	M24217/7-062	M24217/8-062	M24217/7-087	M24217/7-097	UR-150S	-138
	M24217/7-090	-007									
M24217/8-021	MS17804-16-16	———	8 NO.16	M24217/8-031	M24217/8-086	M24217/8-098	———	M24217/8-092	M24217/8-105	UR-81S	-119
M24217/8-022	MS17804-16-16	———	14 NO.16 (SPECIAL)	M24217/8-042	M24217/7-042	M24217/7-053	———	M24217/7-081	M24217/7-096	UR-112S	-132

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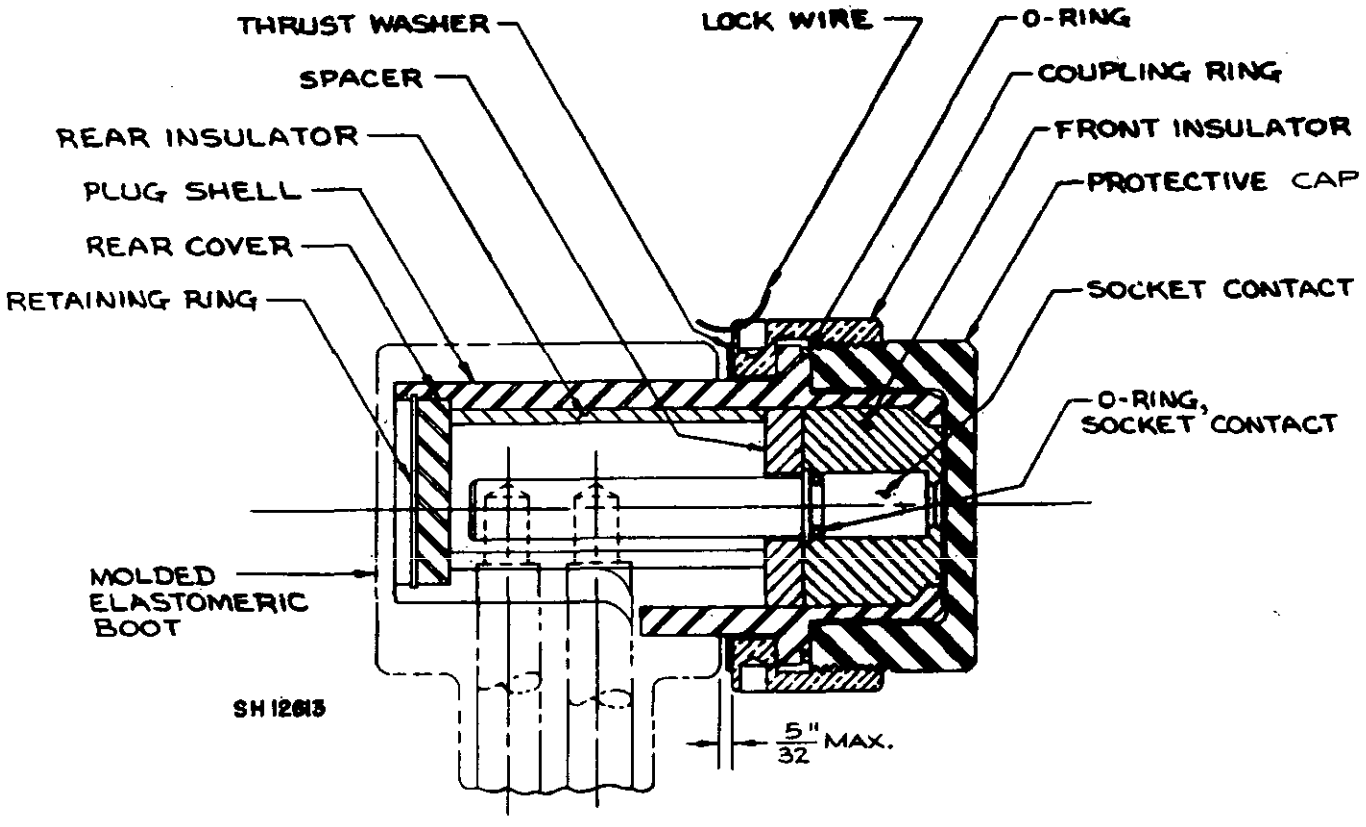


FIGURE 2. Configuration, type II.

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TABLE II. Part numbers for right angle plug assemblies.

PLUG ASSEMBLY NUMBER M24217/	SOCKET CONTACT M24217/	O-RING SOCKET CONTACT AS-568A	NUMBER AND SIZE OF CONTACTS	PLUG SHELL NUMBER M24217/	COUPLING RING NUMBER M24217/	FRONT INSULATOR NUMBER M24217/	REAR INSULATOR NUMBER M24217/	PROT CAP NUMBER M24217/	THRUST WASHER NUMBER M24217/	RET. RING SPIRO-LOX NO.	SHELL O-RING AS-568A	REAR COVER M24217/	SPACER M24217/
8-025	8-080	-013	2 NO. 0 WITH 4 NO. 12	8-048	7-049	7-063	8-070	7-088	7-098	UR-200S	-145	8-114	8-076
	7-089	-004											
8-026	*8-081	-013	3 NO. 0 STYLE A	8-049	8-087	8-099	8-071	8-093	8-106	UR-231S	-147	8-115	8-075
	*8-082	-013											
8-027	*8-081	-013	3 NO. 0 STYLE B	8-050	8-087	8-099	8-071	8-093	8-106	UR-231S	-147	8-115	8-075
	*8-082	-013											
8-028	*8-081	-013	3 NO. 0 STYLE C	8-051	8-087	8-099	8-071	8-093	8-106	UR-231S	-147	8-115	8-075
	*8-082	-013											
8-029	*8-081	-013	3 NO. 0 STYLE D	8-052	8-087	8-099	8-071	8-093	8-106	UR-231S	-147	8-115	8-075
	*8-082	-013											

* NOTE: TWO SOCKET CONTACTS OF EACH SIZE SHALL BE FURNISHED WITH EACH PLUG ASSEMBLY.

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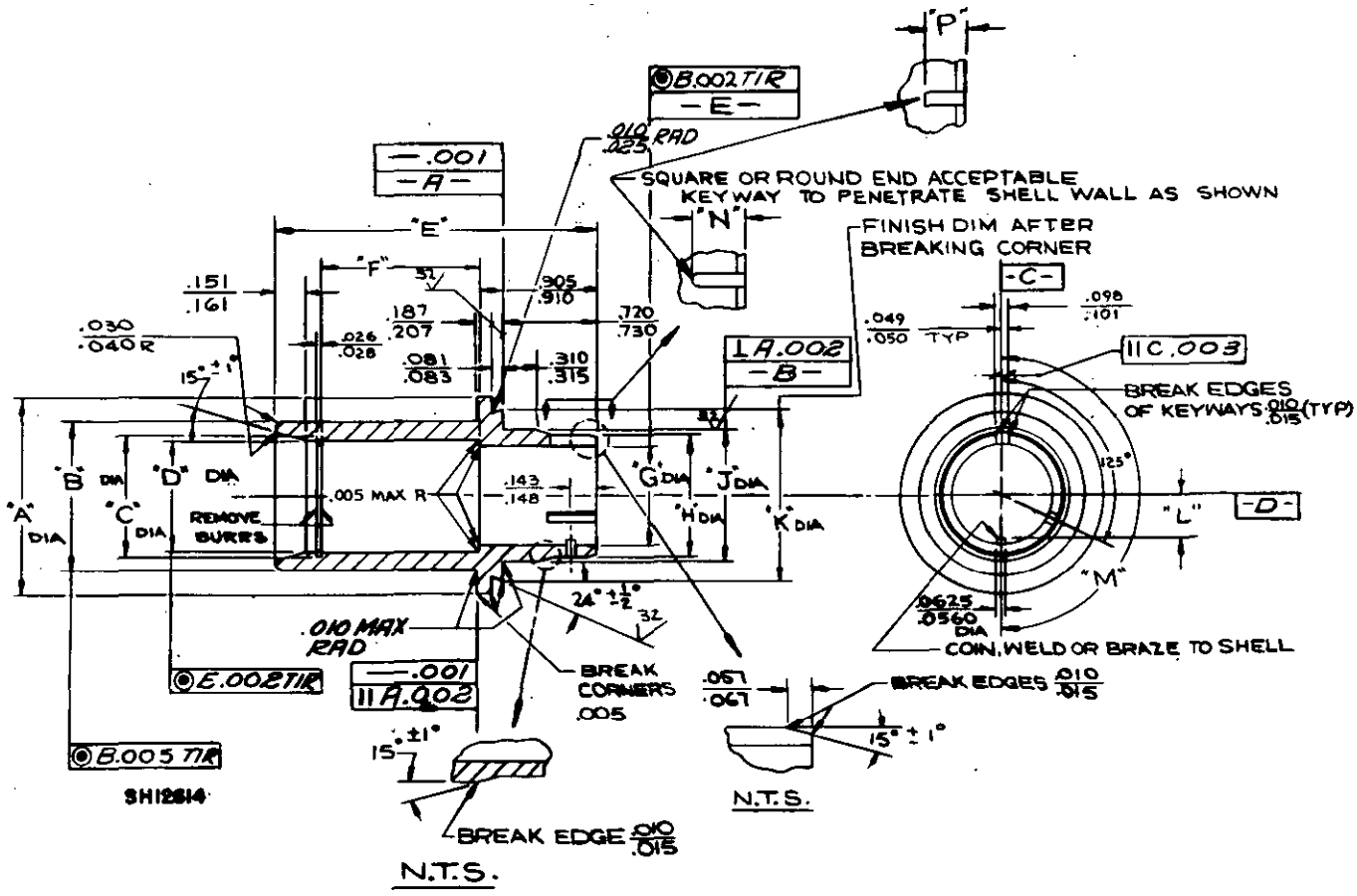


FIGURE 3. Plug shell right angle, type I.

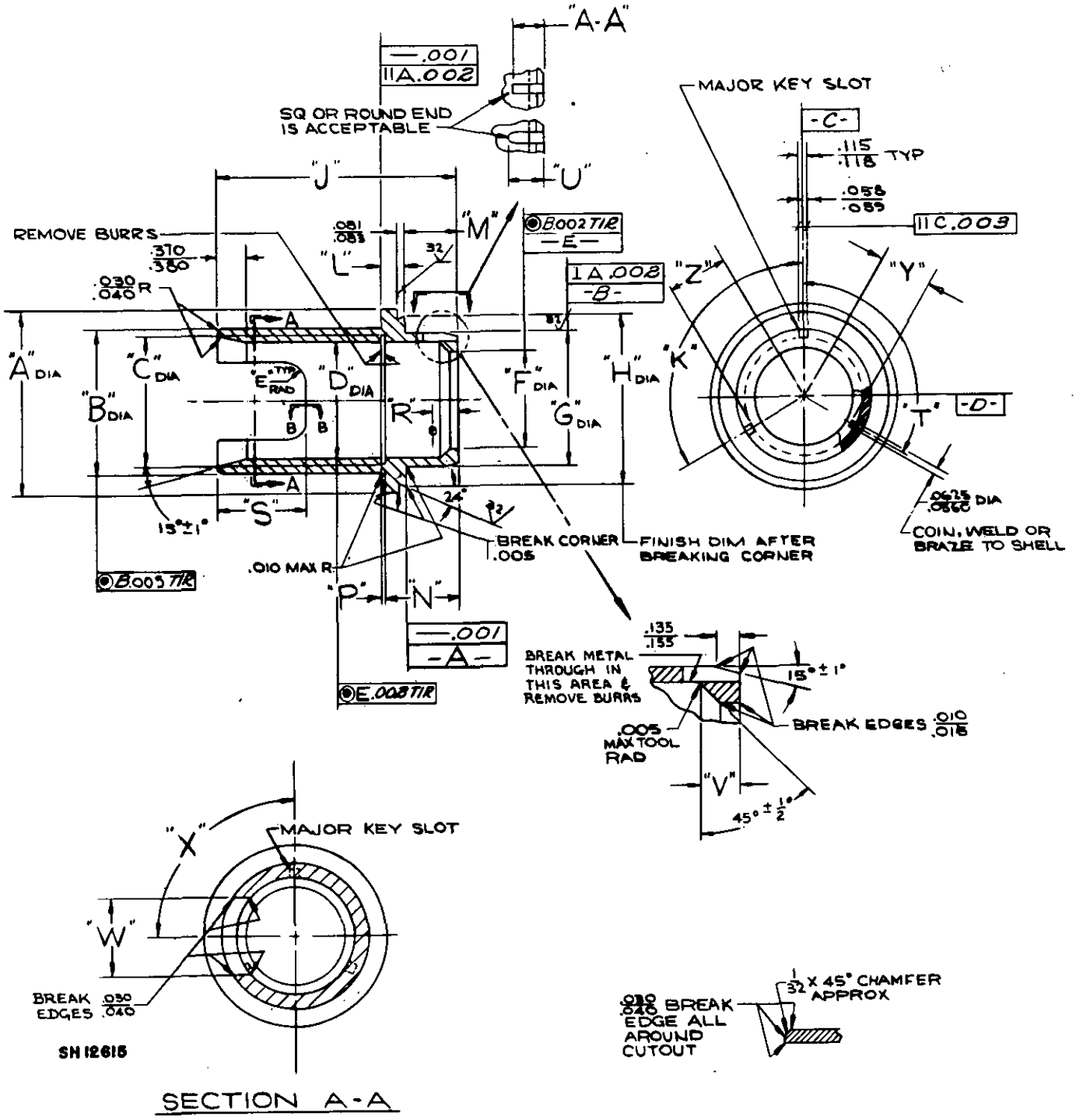
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TABLE III. Dimensions for plug shell, type I.

PLUG SHELL PART NUMBER	NO. & SIZE OF CONTACTS	A	B*	C	D	E	F	G	H	J	K	L	M	N	P
M24217/ 8-031	8 NO. 16	1.255 1.245	1.062 1.060	.851 .857	.813 .817	2.260 2.240	.312 .318	.720 .722	.798 .796	.812 .810	.972 .967	.319 .315	180°	.394 .404	.343 .353

* SEE REQUIREMENT 2

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FOR DIMENSIONS, SEE MIL-C-24217/8 TABLE 2

FIGURE 4. Plug shell, right angle, type II.

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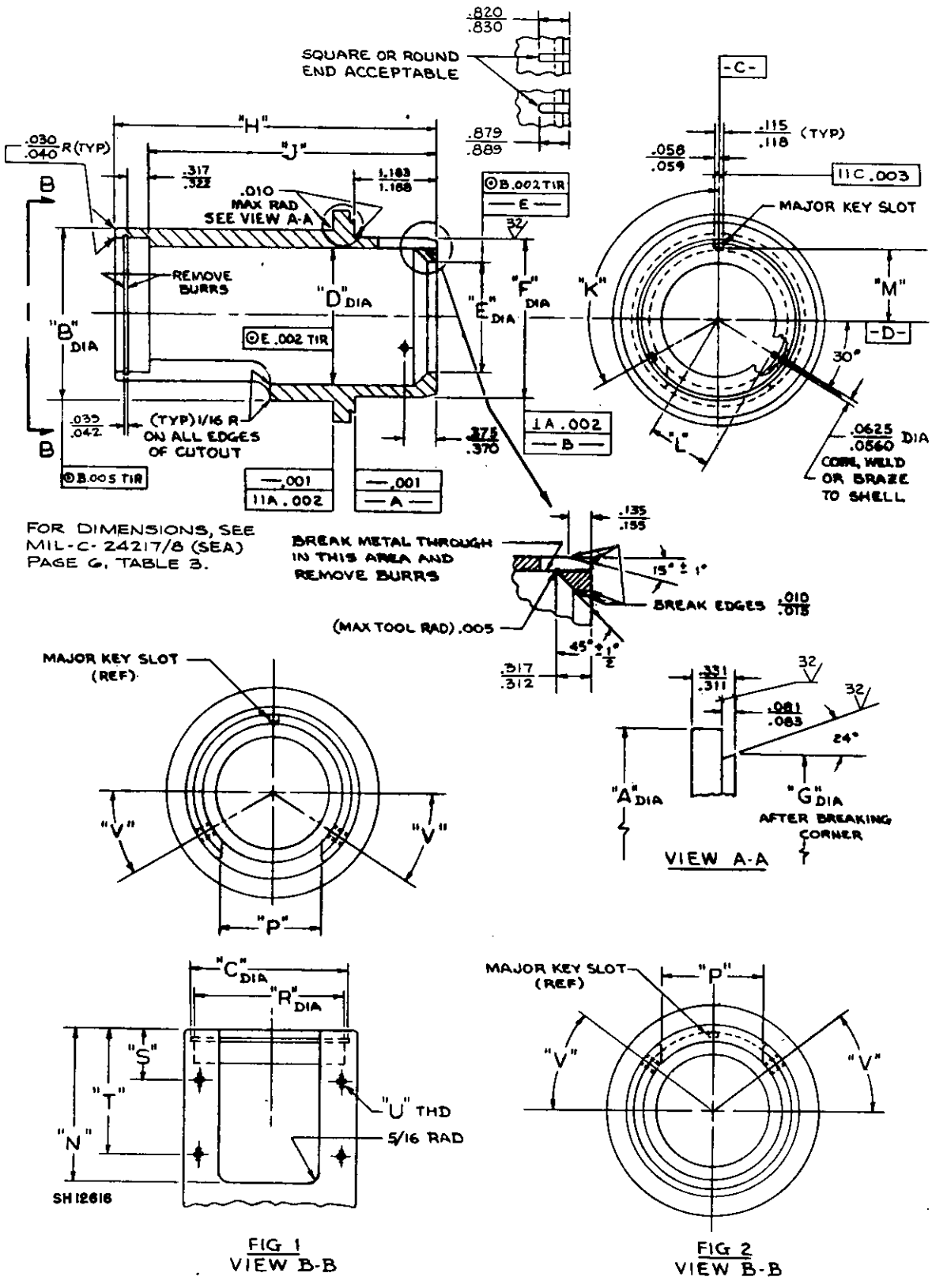


FIG 1
VIEW B-B

FIG 2
VIEW B-B

FIGURE 5. Plug shell, right angle, type III.

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TABLE IV. Dimensions for right angle plug shell, type II.

PLUG SHELL PART NUMBER	NO & SIZE OF CONTACT	A	B*	C	D	E	F	G	H	J	K	L	M	N	P	R	S	T	U	V	W	X	Y	Z	A-A
M24217/8-035	14 NO 16	2.005 1.995	1.502 1.500	1.165 1.173	1.136 1.140	1/8	.954 .956	1.308 1.306	1.792 1.782	3.020 3.000	120°	.269 .249	.720 .730	1.526 1.531	.031 .033	.300 .305	.663 .673	120°	.423 .433	.255 .250	1.010 1.020	90°	.536 .541	.566 .561	.365 .375
M24217/8-036	24 NO 16	2.380 2.370	1.875 1.873	1.540 1.548	1.493 1.497	1/4	1.208 1.210	1.690 1.688	2.167 2.157	3.145 3.125	120°	.269 .249	.720 .730	1.526 1.531	.031 .033	.300 .305	.788 .798	120°	.407 .417	.255 .250	1.396 1.406	90°	.716 .721	.744 .739	.348 .358
M24217/8-037	37 NO 16	2.880 2.870	2.315 2.313	1.864 1.874	1.813 1.817	1/4	1.452 1.454	2.065 2.063	2.605 2.595	3.395 3.375	120°	.331 .311	.720 .730	2.026 2.031	.039 .042	.375 .380	.914 .924	120°	.407 .417	.317 .312	1.732 1.742	180°	.876 .881	.904 .899	.348 .358
M24217/8-038	48 NO 16	3.380 3.360	2.750 2.748	2.320 2.332	2.248 2.252	5/16	2.000 2.002	2.497 2.495	3.047 3.037	3.645 3.625	120°	.340 .320	.995 1.005	2.280 2.285	.039 .042	.300 .305	.727 .737	120°	.681 .691	.255 .250	2.181 2.191	90°	1.094 1.099	1.122 1.117	.622 .632
M24217/8-039	5 NO 12	2.005 1.995	1.502 1.500	1.165 1.173	1.136 1.140	1/4	.954 .956	1.308 1.306	1.792 1.782	3.608 3.588	120°	.269 .249	1.183 1.188	2.500 2.505	.031 .033	.300 .305	.788 .798	120°	.865 .875	.255 .250	1.010 1.020	90°	.536 .541	.566 .561	.806 .816
M24217/8-040	2 NO.12 3 NO.4 3 NO.12 W 3 NO.8	2.380 2.370	1.875 1.873	1.540 1.548	1.493 1.497	1/4	1.208 1.210	1.690 1.688	2.167 2.157	3.608 3.588	120°	.269 .249	1.183 1.188	2.500 2.505	.031 .033	.300 .305	.788 .798	120°	.865 .875	.255 .250	1.396 1.406	90°	.717 .722	.744 .739	.806 .816
M24217/8-041	3 NO 8	2.005 1.995	1.502 1.500	1.165 1.173	1.111 1.115	1/4	.828 .830	1.308 1.306	1.792 1.782	3.450 3.470	120°	.269 .249	1.183 1.188	2.500 2.505	.031 .033	.300 .305	.650 .660	120°	.865 .875	.255 .250	.982 .992	90°	.526 .531	.553 .548	.806 .816
M24217/8-042	14 NO 16	2.005 1.995	1.502 1.500	1.165 1.173	1.136 1.140	1/8	.954 .956	1.308 1.306	1.792 1.782	3.020 3.000	120°	.269 .249	.720 .730	1.526 1.531	.031 .033	.300 .305	.663 .673	120°	.423 .433	.255 .250	1.010 1.020	270°	.536 .541	.566 .561	.365 .375

* SEE REQUIREMENT 2.

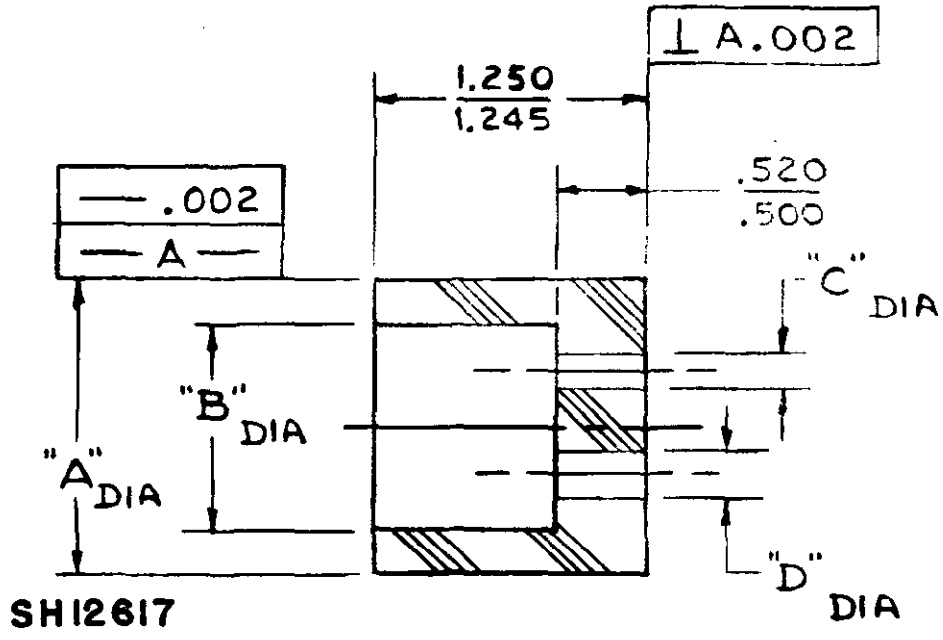
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TABLE V. Dimensions for right angle plug shell, type III.

PLUG SHELL PART NUMBER	NO & SIZE OF CONTACT	A	B*	C	D	E	F	G	H	J	K	L	M	N	P	R	S	T	U	V	CUTOUT LOCATION
M24217/8-048	2 NO 0 4 NO 12	2.880 2.874	2.315 2.313	2.053 2.061	1.813 1.817	1.452 1.454	2.065 2.063	2.600 2.595	5.020 5.000	4.505 4.500	120°	.876 .881	.904 .899	2.125 2.135	1.427 1.437	2.010 2.020	—	—	—	—	FIG 1
M24217/8-049	3 NO 0	3.100 3.090	2.500 2.498	2.382 2.394	2.004 2.008	1.627 1.629	2.312 2.310	2.845 2.840	5.132 5.112	4.619 4.614	120°	.972 .977	1.000 .995	2.240 2.245	1.500 1.510	2.252 2.262	3/4	1 7/8	10-24	35°	FIG 1
M24217/8-050	3 NO 0	3.100 3.090	2.500 2.498	2.382 2.394	2.004 2.008	1.627 1.629	2.312 2.310	2.845 2.840	5.132 5.112	4.619 4.614	150°	.972 .977	1.000 .995	2.240 2.245	1.500 1.510	2.252 2.262	3/4	1 7/8	10-24	35°	FIG 1
M24217/8-051	3 NO 0	3.100 3.090	2.500 2.498	2.382 2.394	2.004 2.008	1.627 1.629	2.312 2.310	2.845 2.840	5.132 5.112	4.619 4.614	120°	.972 .977	1.000 .995	2.240 2.245	1.500 1.510	2.252 2.262	3/4	1 7/8	10-24	35°	FIG 2
M24217/8-052	3 NO 0	3.100 3.090	2.500 2.498	2.382 2.394	2.004 2.008	1.627 1.629	2.312 2.310	2.845 2.840	5.132 5.112	4.619 4.614	150°	.972 .977	1.000 .995	2.240 2.245	1.500 1.510	2.252 2.262	3/4	1 7/8	10-24	35°	FIG 2

* SEE REQUIREMENT 2.

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NOTE: For hole locations, see MIL-C-24217/1, pages 5 and 6.

FIGURE 6. Insulator, rear, type I.

TABLE VI. Dimensions for rear insulator, type I.

REAR INSULATOR PART NUMBER	SIZE/NUMBER OF CONTACTS	A	B	C	D
M24217/8-058	5 NO. 12	1.130 / 1.126	.807 / .817	.180	—
M24217/8-059	9 NO. 12	1.491 / 1.486	1.151 / 1.161	.180	—
M24217/8-060	3 NO. 8	1.109 / 1.104	.807 / .817	.281	—
M24217/8-061	3 NO. 4	1.491 / 1.486	1.120 / 1.130	.406	—
M24217/8-062	3 NO. 8 WITH 3 NO. 12	1.491 / 1.486	1.140 / 1.150	.281	.180

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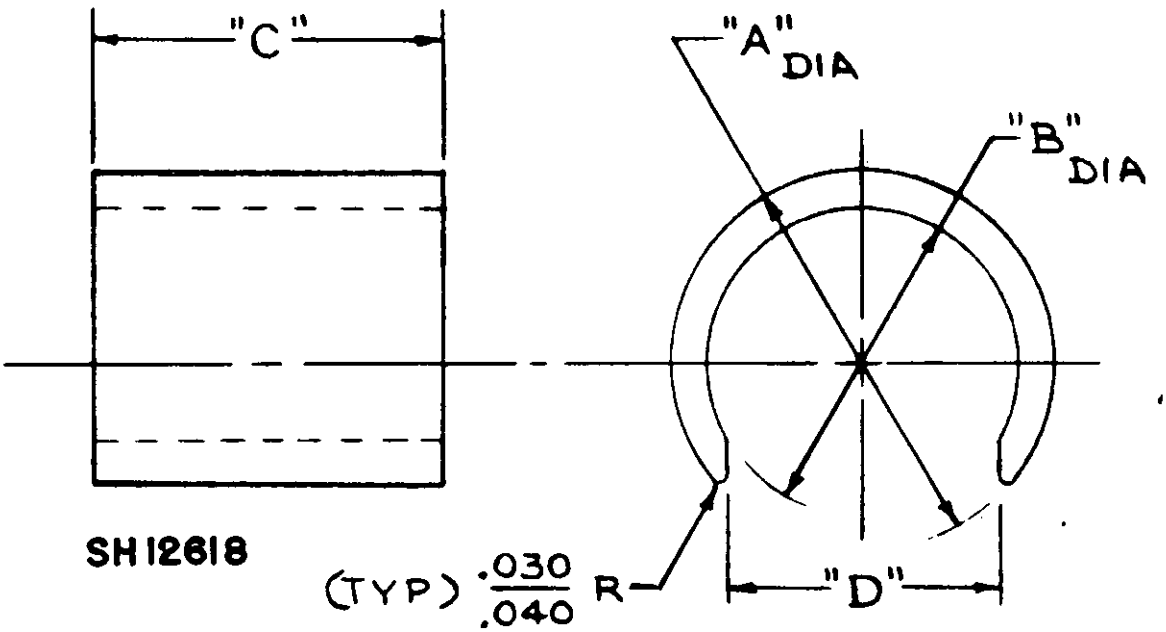


FIGURE 7. Insulator, rear, type II.

TABLE VII. Dimensions for rear insulator, type II.

REAR INSULATOR PART NUMBER	SIZE/NUMBER OF CONTACTS	A	B	C	D
M24217/8-070	2 NO. 0 WITH 4 NO. 12	1.809 / 1.804	1.500 / 1.510	2.178 / 2.173	1.427 / 1.437
M24217/8-071	3 NO. 0	2.000 / 1.995	1.620 / 1.625	1.802 / 1.797	1.500 / 1.510

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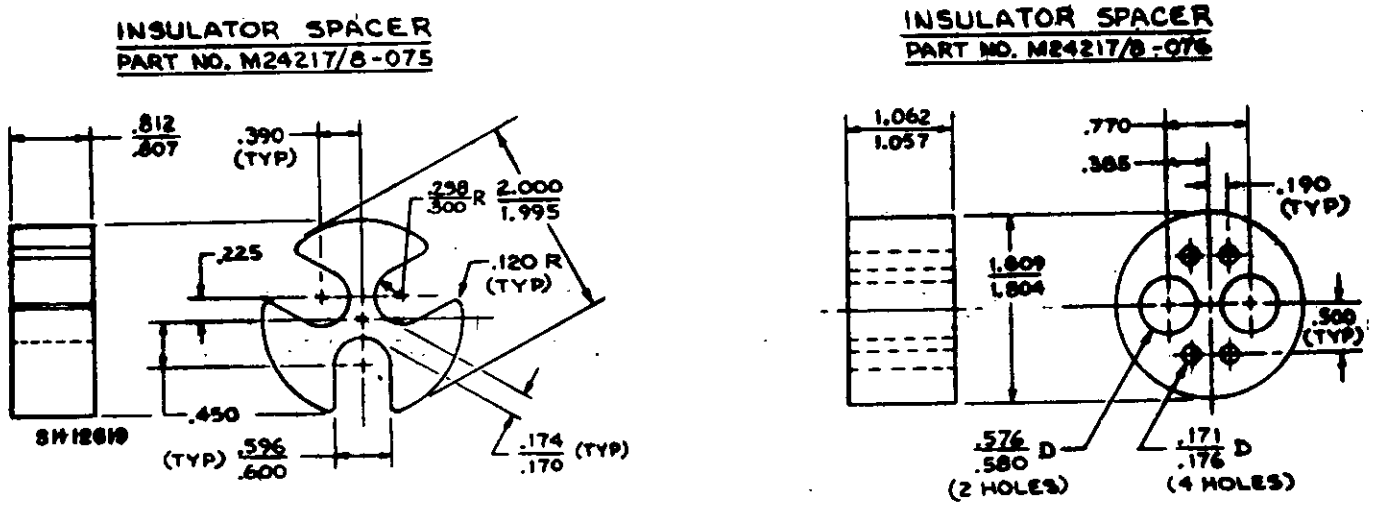
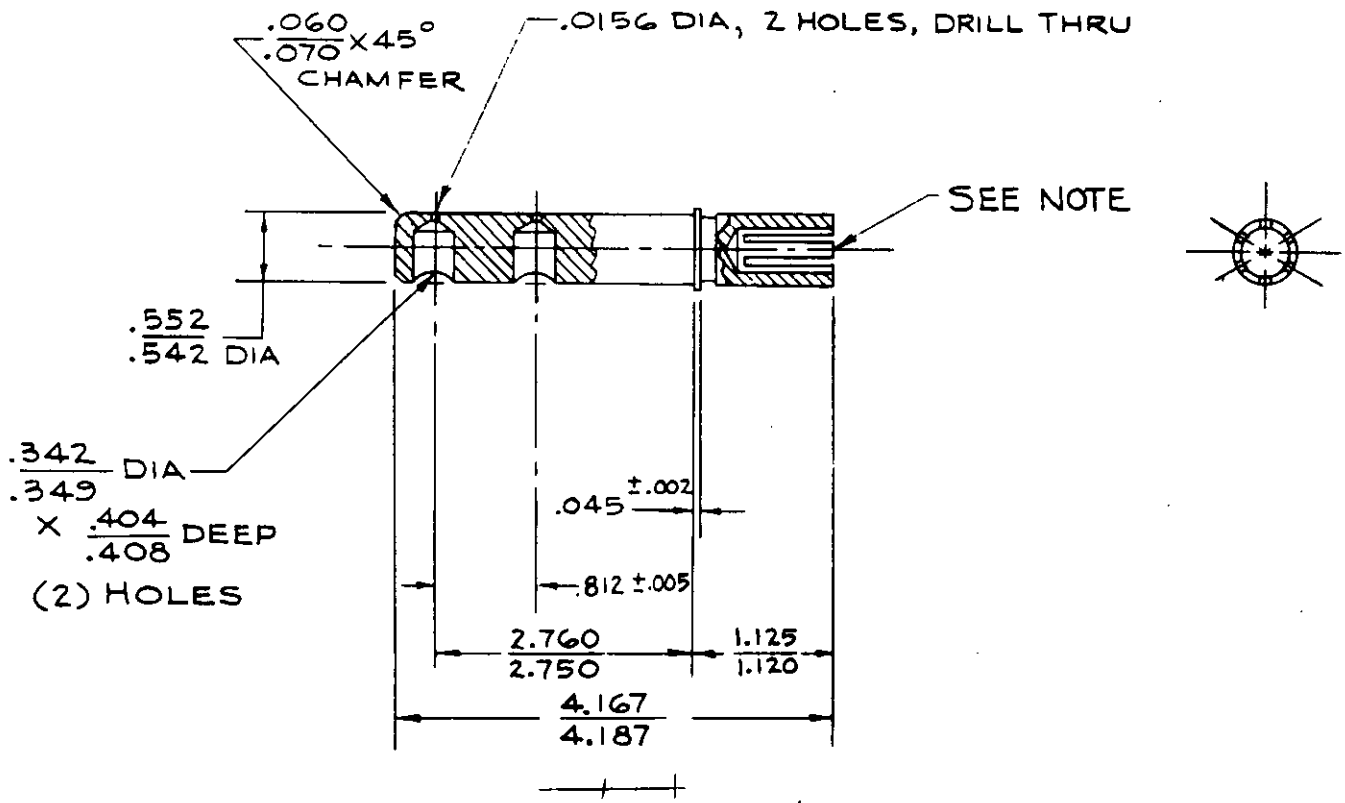


FIGURE 8. Insulator spacers.

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PART NUMBER M24217/8-080

NOTE: FOR ALL OTHER DIMENSIONS SEE DETAIL OF PART NO. M24217/7-092.

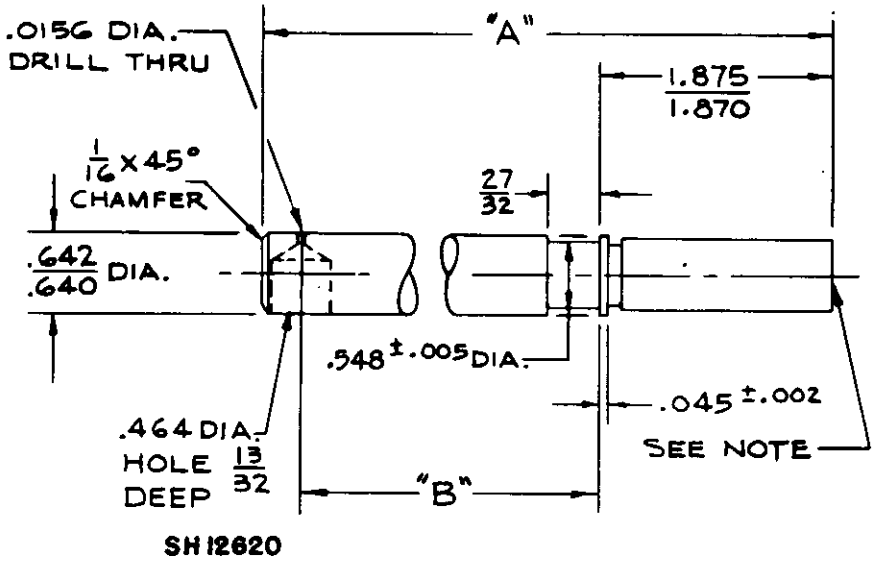


FIGURE 9. Socket, contact.

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TABLE VIII. Dimensions for socket contact.

DIMENSIONS FOR SOCKET CONTACT			
SOCKET CONTACT PART NO.	SIZE	A	B
M24217/8-081	#0	4.247 / 4.242	2.062 / 2.052
M24217/8-082	#0	3.497 / 3.492	1.312 / 1.297

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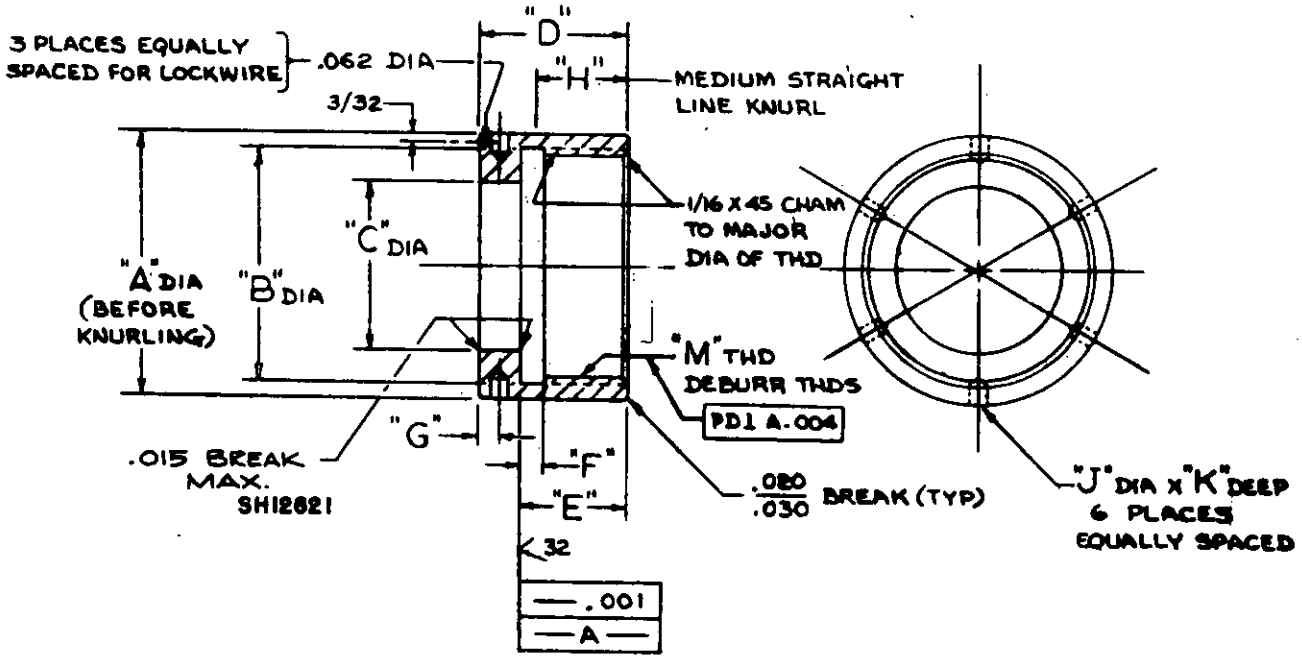


FIGURE 10. Coupling ring.

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TABLE IX. Dimensions for coupling ring.

COUPLING RING PART NUMBER	NUMBER/SIZE OF CONTACTS	A	B	C	D	E	F	G	H	J	K	M
M24217/ 8-086	8 NO. 16	1.510 1.490	1.385 1.395	1.072 1.082	.760 .740	.430 .440	.120 .140	.146 .166	.300 .380	.155 .161	1/8	1 3/8 -12 UN-2B
M24217/ 8-087	3 NO. 0	3.645 3.625	3.260 3.270	2.515 2.520	1.471 1.461	1.028 1.038	.190 .210	.185 .205	.750 .840	.264 .270	1/4	3 1/4 -12 UN-2B

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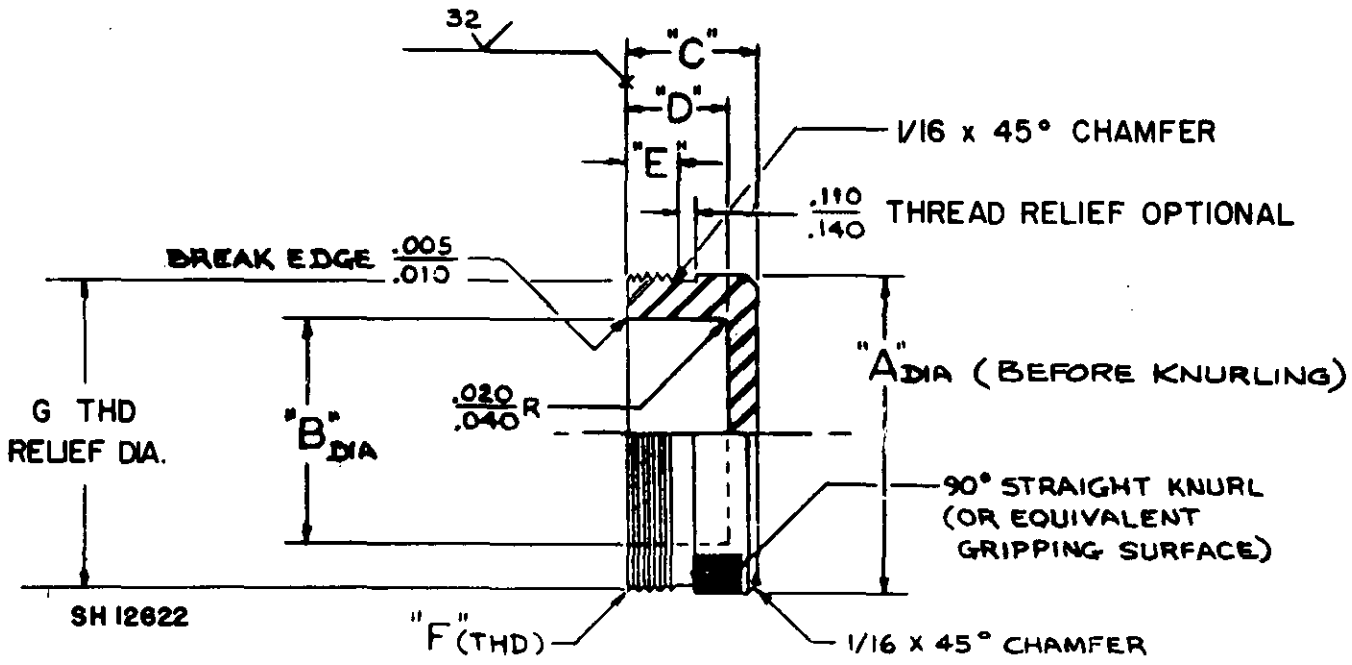


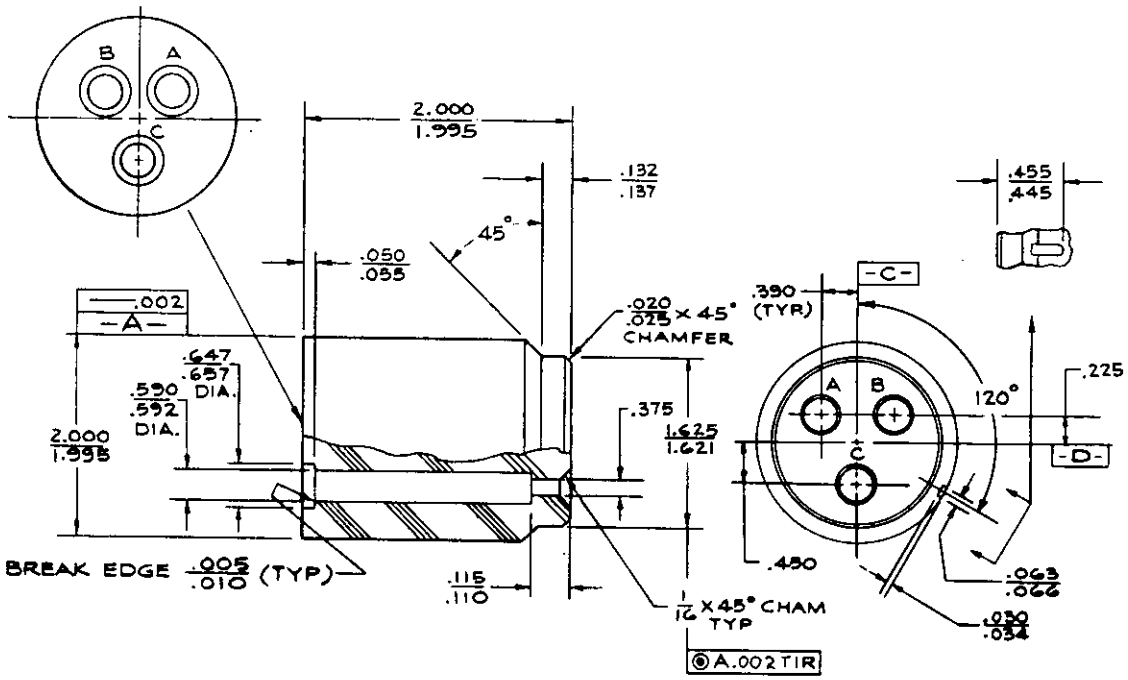
FIGURE 11. Protective cap.

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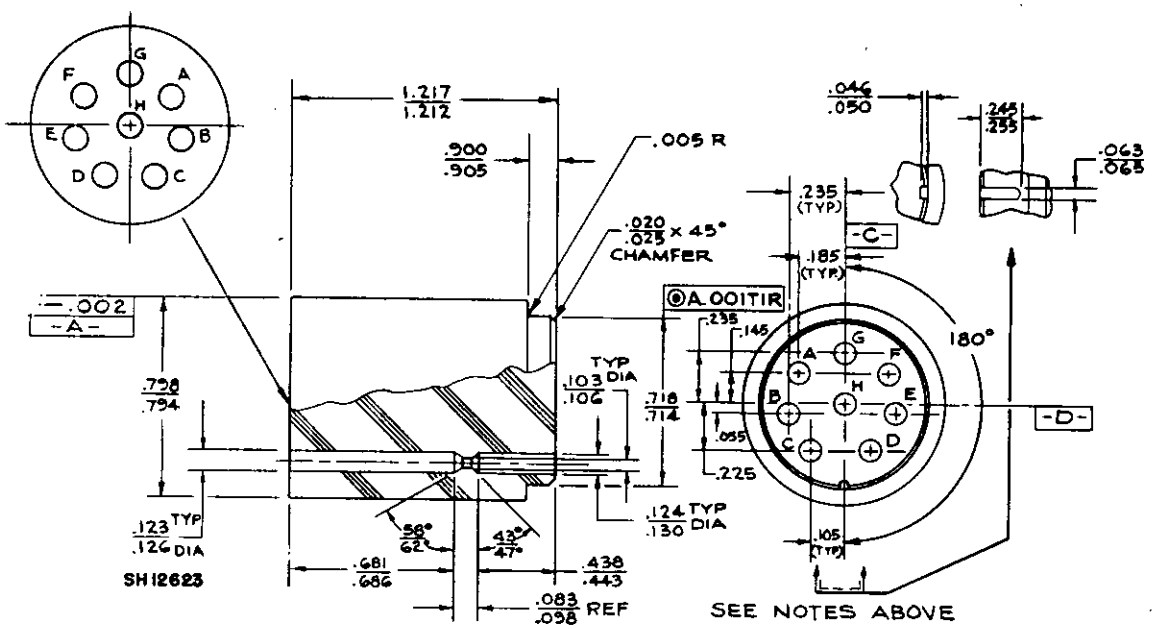
TABLE X. Dimensions for protective cap.

PROT CAP PART NUMBER	NUMBER/SIZE OF CONTACTS	A	B	C	D	E	F (THD)	G
M24217/ 8-092	8 NO. 16	1.380 / 1.360	.875 / .865	1.020 / 1.000	.750 / .770	.350 / .370	$1\frac{3}{8}$ - 12UN-2A	1.26 / 1.24
M24217/ 8-093	3 NO. 0	3.270 / 3.250	2.320 / 2.340	1.485 / 1.463	1.213 / 1.233	.813 / .833	$3\frac{1}{4}$ - 12UN-2A	3.13 / 3.11

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3 NO. 0 - PART NO. M24217/8-099



8 NO. 16 - PART NO. M24217/8-098

NOTES:

1. Datums -C- and -D- are vertical and horizontal centerlines of "A" diameter.
2. Contact holes to be true position within 0.003 diameter to datums -C- and -D-.

FIGURE 12. Insulators, front.

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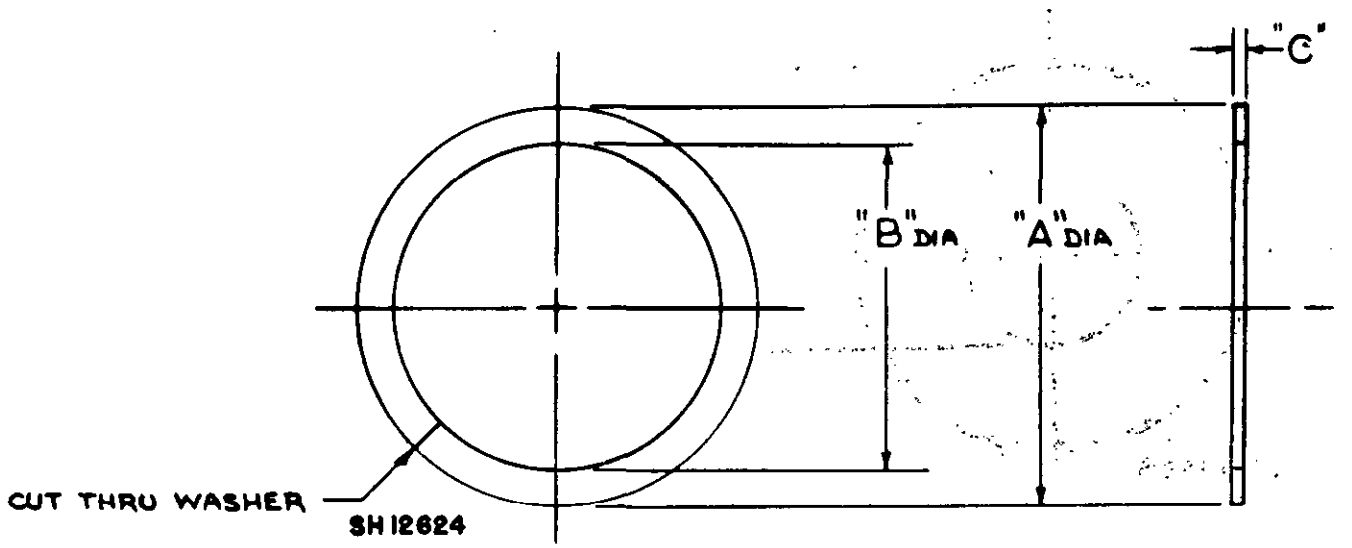


FIGURE 13. Thrust washer.

TABLE XI. Dimensions for thrust washer.

THRUST WASHER PART NUMBER	NUMBER & SIZE OF CONTACTS	A	B	C
M24217/8-105	8 NO. 16	1.390 / 1.370	1.067 / 1.087	.026 / .036
M24217/8-106	3 NO. 0	3.200 / 3.180	2.512 / 2.532	.026 / .036

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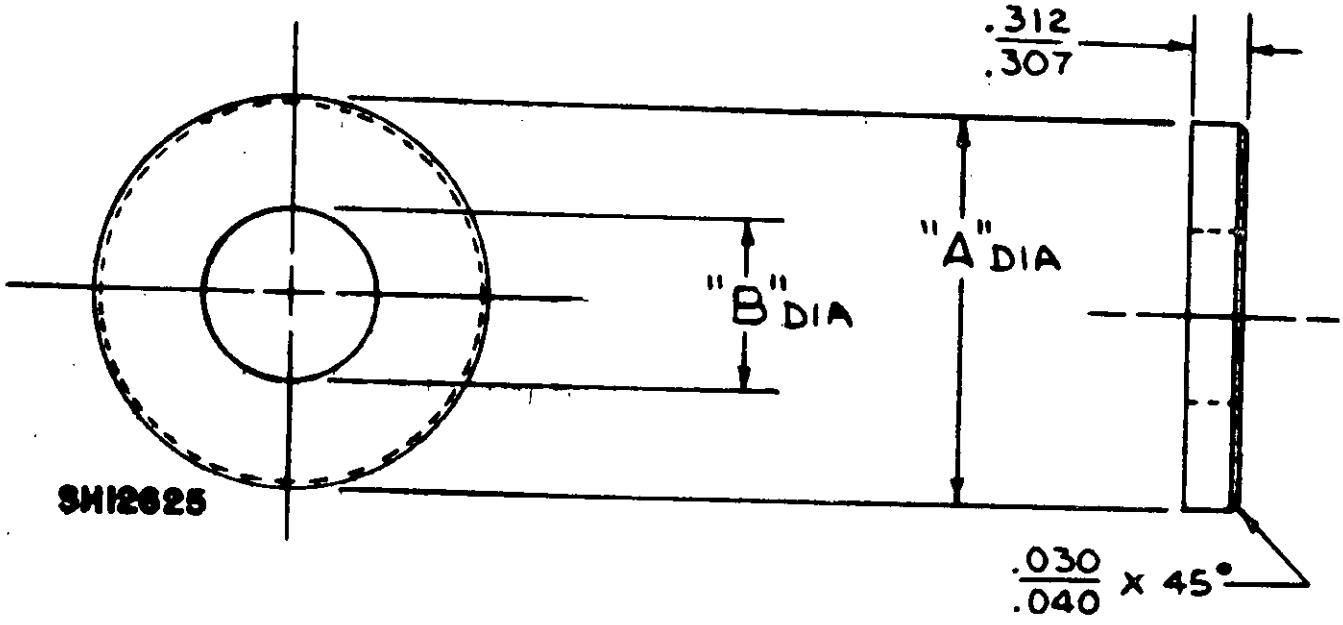


FIGURE 14. Rear cover.

TABLE XII. Dimensions for rear cover.

REAR COVER PART NUMBER	NUMBER & SIZE OF CONTACTS	A	B
M24217/8-114	2 NO. 0 WITH 4 NO. 12	1.995 / 1.990	—
M24217/8-115	3 NO. 0	1.250 / 1.248	1

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REQUIREMENTS:

1. Dimensions and configuration: See figures and tables.
2. Dimensions shown in plug shell "B" column are applicable for that portion of the shell that fits into the mold cavity. Dimensions on the remaining "B" diameter area may be increased or decreased by 0.004 inch as applicable.
3. The material for the spacer, part numbers M24217/8-075 and M24217/8-076 shall be molded glass filled diallyl phthalate, type GDI-30F in accordance with MIL-M-14 or machined from a glass fabric base melamine resin, type GMG in accordance with MIL-P-79.
4. The material for the rear cover, part numbers M24217/8-114 and M24217/8-115 shall be corrosion-resistant steel, class 316 in accordance with QQ-S-763, and shall be passivated.
5. Alternate material for rear insulators, part numbers M24217/8-070 and M24217/8-071 is a glass fabric base melamine resin (tube), type GMG in accordance with MIL-P-79.
6. Spacing between the coupling ring and the molded elastomeric boot shall not exceed 5/32 inch to ensure adequate bonding area.
7. Optional lock wire shall be of nickel copper in accordance with MS20995.
8. Plug shells, for right angle plugs, ordered from another specification sheet shall not be modified to include a cutout.
9. Part number M24217/8-021 shall not be used for new design applications.

Preparing activity:
Navy - SH
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